4 The Space of Meromorphic Functions $L(D)$

Once divisors encode the zeros and poles of meromorphic functions, a natural question is

Which meromorphic functions have zeros and poles compatible with a given divisor?

This leads to the definition

$$L(D) = \{\, f \in \mathcal{M}(X) \mid \operatorname{div}(f) + D \geq 0 \,\} \cup \{0\},$$

the space of meromorphic functions whose poles are bounded by D.

Observation. Let $p \in X$, and let z be a local coordinate with $z(p) = 0$. Any nonzero meromorphic function f near p can be written as

$$f = z^{-m}u, \quad u(p) \neq 0,$$

for some $m \in \mathbb{Z}$. Now suppose we want to allow at most a pole of order n at p. Then

$$f \text{ has pole of order at most } n \text{ at } p \iff z^n f \text{ is holomorphic at } p.$$

Since

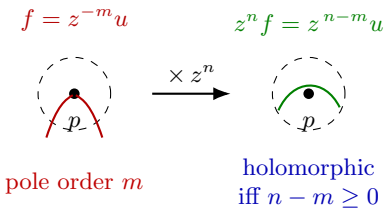
$$\frac{\mathrm{d}(z^n f)}{z^n f} = \frac{f \cdot n z^{n-1}\mathrm{d}z + z^n \cdot f'\mathrm{d}z}{z^n f} = n \cdot \frac{\mathrm{d}z}{z} + \frac{\mathrm{d}f}{f},$$

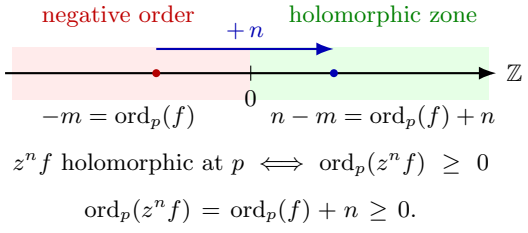
we have

$$\operatorname{ord}_p(z^n f) = \frac{1}{2\pi i}\oint_{\gamma_p} \frac{\mathrm{d}(z^n f)}{z^n f} = \frac{1}{2\pi i}\left(n\oint_{\gamma_p}\frac{\mathrm{d}z}{z} + \oint_{\gamma_p}\frac{\mathrm{d}f}{f}\right) = n + \operatorname{ord}_p(f).$$

Thus,

$$f \text{ has pole order at most } n \text{ at } p \iff z^n f \text{ is holomorphic at } p \iff \operatorname{ord}_p(f) + n \geq 0.$$





Divisor Group

Definition 4. Let X be a compact Riemann surface. The **divisor group** is

$$\mathrm{Div}(X) := \left\{ \sum_{p\in X} n_p[p] \;\middle|\; n_p \in \mathbb{Z}, \text{ and } n_p = 0 \text{ for all but finitely many } p \right\}.$$

The group operation is addition:

$$\left(\sum_p n_p[p]\right) + \left(\sum_p m_p[p]\right) = \sum_p (n_p + m_p)[p].$$

Since addition of integers is commutative, $\mathrm{Div}(X)$ is an abelian group.

Remark 5. Consider the following isomorphism

$$\begin{array}{rccc} \Phi & : & \mathrm{Div}(X) & \longrightarrow & \bigoplus_{p\in X} \mathbb{Z} \\ & & \sum_{p\in X} n_p[p] & \longmapsto & (n_p)_{p\in X} \end{array}.$$

That is, $\mathrm{Div}(X) \cong \bigoplus_{p\in X} \mathbb{Z}$, so $\mathrm{Div}(X)$ is the free abelian group generated by the points of X.

Remark 6. A divisor $D = \sum_p n_p[p]$ is called **effective** if $n_p \geq 0$ for all $p \in X$, and we write $D \geq 0$.

Space of Meromorphic Functions (Riemann–Roch Space)

Definition 5. Let

$$D = \sum_{p\in\mathbb{CP}^1} n_p[p] \in \mathrm{Div}(\mathbb{CP}^1)$$

be a divisor on $\mathbb{CP}^1$. The **space of meromorphic functions associated to** D is

$$L(D) := \{\, f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \mathrm{div}(f) + D \geq 0 \,\} \cup \{0\}.$$

Equivalently,

$$L(D) = \{\, f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \mathrm{ord}_p(f) \geq -n_p \text{ for every } p \in \mathbb{CP}^1 \,\} \cup \{0\}.$$

Its dimension is denoted by

$$\ell(D) := \dim_{\mathbb{C}} L(D).$$

Remark 7. For a general divisor $D = \sum_{p \in X} n_p[p]$ on a compact Riemann surface X,

$$f \in L(D) \iff \operatorname{ord}_p(f) + n_p \geq 0 \text{ for all } p \in X.$$

Therefore

- if $n_p > 0$, then f may have a pole at p, but of order at most n_p;
- if $n_p = 0$, then f cannot have a pole at p;
- if $n_p < 0$, then f must have a zero at p of order at least $-n_p$.

Remark 8. Take $D = \sum_{p \in X} n_p[p]$ and consider $L(D)$. Then

(i) By definition, $0 \in L(D)$.

(ii) Let $f \in L(D)$ and $c \in \mathbb{C}$. If $c = 0$, then $cf = 0 \in L(D)$. If $c \neq 0$, then

$$\operatorname{ord}_p(cf) = \frac{1}{2\pi i} \oint_{\gamma_p} \frac{\mathrm{d}(cf)}{cf} = \frac{1}{2\pi i} \oint_{\gamma_p} \frac{\mathrm{d}f}{f} = \operatorname{ord}_p(f).$$

for every point $p \in X$. Then

$$\operatorname{ord}_p(f) + n_p \geq 0 \implies \operatorname{ord}_p(cf) + n_p \geq 0$$

for all $p \in X$, hence $cf \in L(D)$.

(iii) Let $f, g \in L(D)$ with $f + g \neq 0$. Fix a point $p \in X$. Define

$$m := \min\{\operatorname{ord}_p(f), \operatorname{ord}_p(g)\}.$$

Choose a local coordinate z centered at p. Then

$$f = z^m u, \qquad g = z^m v,$$

where u and v are holomorphic near p, and at least one of $u(p), v(p)$ is nonzero. Then

$$\operatorname{ord}_p(f + g) = \operatorname{ord}_p(z^m(u + v)) = m + \operatorname{ord}_p(u + v).$$

Since $u + v$ is holomorphic ($\operatorname{ord}_p(u + v) \geq 0$), we have

$$\operatorname{ord}_p(f + g) \geq m = \min\{\operatorname{ord}_p(f), \operatorname{ord}_p(g)\}.$$

Since $f, g \in L(D)$, we have

$$\mathrm{ord}_p(f) + n_p \geq 0 \quad \text{and} \quad \mathrm{ord}_p(g) + n_p \geq 0,$$

so

$$m + n_p = \min\{\mathrm{ord}_p(f), \mathrm{ord}_p(g)\} + n_p \geq 0.$$

Therefore

$$\mathrm{ord}_p(f+g) + n_p \geq m + n_p \geq 0,$$

so $f + g \in L(D)$.

Hence $L(D)$ is a vector space over $\mathbb{C}$. If X is compact, this vector space is in fact finite-dimensional, and we write

$$\ell(D) := \dim_{\mathbb{C}} L(D).$$

Example 4 (Computing $L(D)$ and $\ell(D)$)**.**

(1) Let $D = 0$ (zero divisor). Then

$$L(0) = \{\, f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \mathrm{div}(f) \geq 0 \,\} \cup \{0\} = \{\, f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \mathrm{ord}_p(f) \geq 0 \text{ for all } p \in \mathbb{CP}^1 \} \cup \{0\}.$$

So f has no poles anywhere on $\mathbb{CP}^1$. Hence f is holomorphic everywhere on the compact Riemann surface $\mathbb{CP}^1$, so f must be constant. Therefore

$$L(0) = \mathbb{C} \quad \text{and} \quad \ell(0) = \dim_{\mathbb{C}} L(0) = 1.$$

(2) Let $D = [0]$ (simple zero at 0). Then

$$\begin{aligned} L([0]) &= \{\, f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \mathrm{div}(f) + [0] \geq 0 \,\} \cup \{0\} \\ &= \{\, f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \textstyle\sum_{p \in \mathbb{C}\setminus\{0\}} \mathrm{ord}_p[p] + (\mathrm{ord}_0(f) + 1)[0] + (\mathrm{ord}_\infty(f))[\infty] \geq 0 \,\} \cup \{0\} \\ &= \{\, f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \mathrm{ord}_{p \notin \{0,\infty\}}(f) \geq 0;\ \mathrm{ord}_0(f) \geq -1;\ \mathrm{ord}_\infty(f) \geq 0 \} \cup \{0\} \\ &= \left\{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \begin{array}{c} f \text{ has no pole at } \infty \\ f \text{ may have pole of order} \leq 1 \text{ at } 0 \end{array} \right\} \cup \{0\} \end{aligned}$$

Let

$$f(z) = \frac{P(z)}{z}, \quad P(z) \in \mathbb{C}[z].$$

Note that

$$\begin{aligned} P(z) &= a_0 + a_1 z + \cdots + a_{\deg P} z^{\deg P} \quad (a_{\deg P} \neq 0), \\ P(1/w) &= a_0 + a_1 w^{-1} + \cdots + a_{\deg P} w^{-\deg P}. \end{aligned}$$

Since no pole is allowed at ∞, we require

$$\mathrm{ord}_\infty(f) = \mathrm{ord}_{w=0}(wP(1/w)) = 1 - \deg P \geq 0,$$

so $\deg P \leq 1$. Thus $P(z) = az + b$ $(a, b \in \mathbb{C})$ and therefore

$$f(z) = \frac{az+b}{z} = a + \frac{b}{z}.$$

Hence

$$L([0]) = \left\{ a + \frac{b}{z} : a, b \in \mathbb{C} \right\} = \mathrm{span}_{\mathbb{C}} \left\{ 1, \frac{1}{z} \right\} \quad \text{so} \quad \ell([0]) = 2.$$

(3) Let $D = [0] + [1]$. Then

$$\begin{aligned}
&L([0]+[1]) \\
&= \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \mathrm{div}(f) + [0] + [1] \geq 0 \} \cup \{0\} \\
&= \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \textstyle\sum_{p \in \mathbb{C}\setminus\{0,1\}} \mathrm{ord}_p[p] + (\mathrm{ord}_0(f)+1)[0] + (\mathrm{ord}_1(f)+1)[1] + (\mathrm{ord}_\infty(f))[\infty] \geq 0 \} \cup \{0\} \\
&= \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \mathrm{ord}_{p \notin \{0,\infty\}}(f) \geq 0;\ \mathrm{ord}_0(f) \geq -1;\ \mathrm{ord}_1(f) \geq -1;\ \mathrm{ord}_\infty(f) \geq 0 \} \cup \{0\} \\
&= \left\{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \begin{array}{c} f \text{ has no pole at } \infty \\ f \text{ may have pole of order} \leq 1 \text{ at } 0 \\ f \text{ may have pole of order} \leq 1 \text{ at } 1 \end{array} \right\} \cup \{0\}
\end{aligned}$$

Let

$$f(z) = \frac{P(z)}{z(z-1)}, \quad P(z) \in \mathbb{C}[z].$$

Since no pole is allowed at ∞, we require

$$\mathrm{ord}_\infty(f) = \mathrm{ord}_{w=0}\left(w \cdot \frac{w}{1-w} \cdot P(1/w) \right) = \mathrm{ord}_{w=0}\left(w^2 \cdot \frac{1}{1-w} \cdot P(1/w) \right) = 2 - \deg P \geq 0,$$

so $\deg P \leq 2$. Thus $P(z) = az^2 + bz + c$ $(a, b, c \in \mathbb{C})$ and therefore

$$f(z) = \frac{az^2+bz+c}{z(z-1)} = A + \frac{B}{z} + \frac{C}{z-1}.$$

Since

$$az^2 + bz + c = Az(z-1) + B(z-1) + Cz = Az^2 + (-A+B+C)z - B,$$

we have

$$\begin{cases} A = a \\ -A + B + C = b \\ -B = c \end{cases} \implies \begin{cases} A = a \\ B = -c \\ C = a + b + c \end{cases}.$$

Thus

$$f(z) = a - \frac{c}{z} + \frac{a+b+c}{z-1}, \qquad a, b, c \in \mathbb{C}.$$

Therefore

$$L([0]+[1]) = \left\{ a - \frac{c}{z} + \frac{a+b+c}{z-1} : a, b, c \in \mathbb{C} \right\} = \operatorname{span}\left\{ 1, \frac{1}{z}, \frac{1}{z-1} \right\} \quad \text{and} \quad \ell([0]+[1]) = 3.$$

(4) Let $D = [\infty]$. Then

$$\begin{aligned} L([\infty]) &= \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \operatorname{div}(f) + [\infty] \geq 0 \} \cup \{0\} \\ &= \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \textstyle\sum_{p \in \mathbb{C}} \operatorname{ord}_p[p] + (\operatorname{ord}_\infty(f)+1)[\infty] \geq 0 \} \cup \{0\} \\ &= \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \operatorname{ord}_{p \in \mathbb{C}}(f) \geq 0;\ \operatorname{ord}_\infty(f) \geq -1 \} \cup \{0\} \\ &= \left\{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \begin{array}{c} f \text{ may have a pole of order} \leq 1 \text{ at } \infty \\ f \text{ has no pole at any finite point } p \in \mathbb{C} \end{array} \right\} \cup \{0\} \end{aligned}$$

Let

$$f(z) = P(z) \quad \text{with} \quad P(z) \in \mathbb{C}[z].$$

Since f may have a pole of order at most 1 at ∞, we require

$$\operatorname{ord}_\infty(f) = \operatorname{ord}_{w=0}\left(P(1/w)\right) = -\deg P \geq -1,$$

so $\deg P \leq 1$. Thus $P(z) = az + b$ $(a, b \in \mathbb{C})$ and therefore

$$f(z) = az + b, \quad a, b \in \mathbb{C}.$$

Hence

$$L([\infty]) = \{ az + b : a, b \in \mathbb{C} \} = \operatorname{span}_{\mathbb{C}}\{1, z\} \quad \text{and} \quad \ell([\infty]) = 2.$$

(5) Let $D = 2[\infty]$. Then

$$L(2[\infty]) = \{ f \in \mathcal{M}(\mathbb{CP}^1) \mid \operatorname{div}(f) + 2[\infty] \geq 0 \} \cup \{0\}.$$

Thus f has no poles on $\mathbb{C}$ and at most a double pole at ∞. Equivalently, f is a polynomial of degree at most 2. Therefore

$$L(2[\infty]) = \left\{ az^2 + bz + c : a, b, c \in \mathbb{C} \right\} = \operatorname{span}_{\mathbb{C}}\{1, z, z^2\} \quad \text{and} \quad \ell(2[\infty]) = 3.$$

(6) Let $D = [0] + [\infty]$. Then

$$\begin{aligned} L([0]+[\infty]) &= \{\, f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \operatorname{div}(f) + [0] + [\infty] \geq 0 \,\} \cup \{0\} \\ &= \{\, f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \textstyle\sum_{p\in\mathbb{C}\setminus\{0\}} \operatorname{ord}_p[p] + (\operatorname{ord}_0(f)+1)[0] + (\operatorname{ord}_\infty(f)+1)[\infty] \geq 0 \,\} \cup \{0\} \\ &= \{\, f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \operatorname{ord}_{p\in\mathbb{C}\setminus\{0\}}(f) \geq 0;\ \operatorname{ord}_0(f) \geq -1;\ \operatorname{ord}_\infty(f) \geq -1 \} \cup \{0\} \\ &= \left\{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \begin{matrix} f \text{ may have a pole of order} \leq 1 \text{ at } \infty \\ f \text{ may have a pole of order} \leq 1 \text{ at } 0 \end{matrix} \right\} \cup \{0\} \end{aligned}$$

Let

$$f(z) = \frac{P(z)}{z} \quad \text{with} \quad P(z) \in \mathbb{C}[z].$$

Since f may have a pole of order at most 1 at ∞, we require

$$\operatorname{ord}_\infty(f) = \operatorname{ord}_{w=0}\left(wP(1/w)\right) = 1 - \deg P \geq -1,$$

so $\deg P \leq 2$. Thus $P(z) = az^2 + bz + c$ $(a, b, c \in \mathbb{C})$ and therefore

$$f(z) = \frac{az^2+bz+c}{z} = az + b + \frac{c}{z}, \quad a, b, c \in \mathbb{C}.$$

Hence

$$L([0]+[\infty]) = \left\{ az + b + c/z : a, b, c \in \mathbb{C} \right\} = \operatorname{span}\left\{1, z, \frac{1}{z}\right\} \quad \text{and} \quad \ell([0]+[\infty]) = 3.$$

(7) Let $D = [\infty] - [1]$. Then

$$\begin{aligned} L([\infty]-[1]) &= \{\, f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \operatorname{div}(f) + [\infty] - [1] \geq 0 \,\} \cup \{0\} \\ &= \{\, f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \textstyle\sum_{p\in\mathbb{C}\setminus\{0\}} \operatorname{ord}_p[p] + (\operatorname{ord}_0(f)-1)[0] + (\operatorname{ord}_\infty(f)+1)[\infty] \geq 0 \,\} \cup \{0\} \\ &= \{\, f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \operatorname{ord}_{p\in\mathbb{C}\setminus\{0\}}(f) \geq 0;\ \operatorname{ord}_0(f) \geq 1;\ \operatorname{ord}_\infty(f) \geq -1 \} \cup \{0\} \\ &= \left\{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \begin{matrix} f \text{ may have a pole of order at most 1 at } \infty \\ f \text{ has a zero of order at least 1 at 1} \end{matrix} \right\} \cup \{0\} \end{aligned}$$

Let

$$f(z) = (z-1)P(z) \quad \text{with} \quad P(z) \in \mathbb{C}[z].$$

Since f may have a pole of order at most 1 at ∞, we require

$$\mathrm{ord}_\infty(f) = \mathrm{ord}_{w=0}\left(\frac{1-w}{w}\cdot P(1/w)\right) = -1 - \deg P \geq -1,$$

so $\deg P \leq 0$. Thus $P(z) = a$ $(a \in \mathbb{C})$ and therefore

$$f(z) = a(z-1), \quad a \in \mathbb{C}.$$

Hence

$$L([\infty]-[1]) = \{a(z-1) : a \in \mathbb{C}\} = \mathrm{span}_{\mathbb{C}}\{z-1\} \quad \text{and} \quad \ell([\infty]-[1]) = 1.$$

(8) Let $D = 2[\infty]-[1]$. Then

$$L(2[\infty]-[1]) = \left\{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \begin{array}{c} f \text{ may have a pole of order at most 2 at } \infty \\ f \text{ has a zero of order at least 1 at 1} \end{array} \right\} \cup \{0\}$$

Let

$$f(z) = (z-1)P(z) \quad \text{with} \quad P(z) \in \mathbb{C}[z].$$

Since f may have a pole of order at most 2 at ∞, we require

$$\mathrm{ord}_\infty(f) = \mathrm{ord}_{w=0}\left(\frac{1-w}{w}\cdot P(1/w)\right) = -1 - \deg P \geq -2,$$

so $\deg P \leq 1$. Thus $P(z) = az + b$ $(a, b \in \mathbb{C})$ and therefore

$$f(z) = (z-1)(az+b) = az^2 + (-a+b)z + b, \quad a, b \in \mathbb{C}.$$

Then

$$L(2[\infty]-[1]) \subseteq \mathrm{span}_{\mathbb{C}}\{1, z, z^2\} \quad \text{but} \quad L(2[\infty]-1) \neq \mathrm{span}_{\mathbb{C}}\{1, z, z^2\}.$$

since $L(2[\infty]-[1])$ has the condition that the function must must be vanish at $z = 1$. Thus,

$$f(z) = (z-1)(az+b) = az(z-1) + b(z-1), \quad a, b \in \mathbb{C}.$$

Hence

$$L(2[\infty]-[1]) = \{az(z-1) + b(z-1) : a, b \in \mathbb{C}\} = \mathrm{span}_{\mathbb{C}}\{z-1, z(z-1)\} \quad \text{and} \quad \ell(2[\infty]-[1]) = 2.$$

(9) Let $D = 2[\infty] - [0] - [1]$. Then

$$
\begin{aligned}
&L(2[\infty] - [0] - [1]) \\
&= \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \operatorname{div}(f) + 2[\infty] - [0] - [1] \geq 0 \} \cup \{0\} \\
&= \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \textstyle\sum_{p \in \mathbb{C}\setminus\{0,1\}} \operatorname{ord}_p[p] + (\operatorname{ord}_0(f) - 1)[0] + (\operatorname{ord}_1(f) - 1)[1] + (\operatorname{ord}_\infty(f) + 2)[\infty] \geq 0 \} \cup \{0\} \\
&= \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \operatorname{ord}_{p \in \mathbb{C}\setminus\{0,1\}}(f) \geq 0;\ \operatorname{ord}_0(f) \geq 1;\ \operatorname{ord}_1(f) \geq 1;\ \operatorname{ord}_\infty(f) \geq -2 \} \cup \{0\} \\
&= \left\{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \begin{array}{c} f \text{ may have a pole of order at most 2 at } \infty \\ f \text{ has a zero of order at least 1 at } 0 \\ f \text{ has a zero of order at least 1 at } 1 \end{array} \right\} \cup \{0\}
\end{aligned}
$$

Let

$$f(z) = z(z-1)P(z) \quad \text{with} \quad P(z) \in \mathbb{C}[z].$$

Since f may have a pole of order at most 2 at ∞, we require

$$\operatorname{ord}_\infty(f) = \operatorname{ord}_{w=0}\left(\frac{1}{w} \cdot \frac{1-w}{w} \cdot P(1/w) \right) = -2 - \deg P \geq -2,$$

so $\deg P \leq 0$. Thus $P(z) = a$ ($a \in \mathbb{C}$) and therefore $f(z) = az(z-1)$ ($a \in \mathbb{C}$). Hence

$$L(2[\infty] - [0] - [1]) = \{az(z-1) : a \in \mathbb{C}\} = \operatorname{span}_{\mathbb{C}}\{z(z-1)\} \quad \text{and} \quad \ell(2[\infty] - [0] - [1]) = 1.$$

(10) Let $D = [0] - [\infty]$. Then

$$
\begin{aligned}
L([0] - [\infty]) &= \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \operatorname{div}(f) + [0] - [\infty] \geq 0 \} \cup \{0\} \\
&= \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \textstyle\sum_{p \in \mathbb{C}\setminus\{0\}} \operatorname{ord}_p[p] + (\operatorname{ord}_0(f) + 1)[0] + (\operatorname{ord}_\infty(f) - 1)[\infty] \geq 0 \} \cup \{0\} \\
&= \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \operatorname{ord}_{p \in \mathbb{C}\setminus\{0\}}(f) \geq 0;\ \operatorname{ord}_0(f) \geq -1;\ \operatorname{ord}_\infty(f) \geq 1 \} \cup \{0\} \\
&= \left\{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \begin{array}{c} f \text{ has a zero of order at least 1 at } \infty \\ f \text{ may have a pole of order at most 1 at } 0 \end{array} \right\} \cup \{0\}
\end{aligned}
$$

Let

$$f(z) = \frac{P(z)}{z} \quad \text{with} \quad P(z) \in \mathbb{C}[z].$$

Since f has a zero of order at least 1 at ∞, we require

$$\operatorname{ord}_\infty(f) = \operatorname{ord}_{w=0}\left(w \cdot P(1/w) \right) = 1 - \deg P \geq 1,$$

so $\deg P \leq 0$. Thus $P(z) = a$ ($a \in \mathbb{C}$) and therefore $f(z) = a/z$ ($a \in \mathbb{C}$). Hence

$$L([0] - [\infty]) = \{a/z : a \in \mathbb{C}\} = \operatorname{span}_{\mathbb{C}}\left\{\frac{1}{z}\right\} \quad \text{and} \quad \ell([0] - [\infty]) = 1.$$

Note. Since every divisor on $\mathbb{CP}^1$ is equivalent to $n[\infty]$, it suffices to compute $L(n[\infty])$.

Proposition 4. *Let $D = n[\infty] \in \mathrm{Div}(\mathbb{CP}^1)$ for some $n \in \mathbb{Z}$. If $n \geq 0$, then*

$$L(n[\infty]) = \mathrm{span}_{\mathbb{C}} \left\{1, z, z^2, \ldots, z^n\right\}.$$

If $n < 0$, then $L(n[\infty]) = 0$. In particular,

$$\ell(n[\infty]) = \begin{cases} n+1, & n \geq 0, \\ 0, & n < 0. \end{cases}$$

Proof. Take a divisor $D = n[\infty] \in \mathrm{Div}(\mathbb{CP}^1)$ with $n \in \mathbb{Z}$. Then

$$\begin{aligned} L(n[\infty]) &= \{\, f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \mathrm{div}(f) + n[\infty] \geq 0 \,\} \cup \{0\} \\ &= \{\, f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \textstyle\sum_{p\in\mathbb{C}} \mathrm{ord}_p[p] + (\mathrm{ord}_\infty(f) + n)[\infty] \geq 0 \,\} \cup \{0\} \\ &= \{\, f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \mathrm{ord}_{p\in\mathbb{C}}(f) \geq 0;\ \mathrm{ord}_\infty(f) \geq -n \} \cup \{0\}. \end{aligned}$$

(i) Assume that $n \geq 0$. Then

$$\begin{aligned} &\{\, f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \mathrm{ord}_{p\in\mathbb{C}}(f) \geq 0;\ \mathrm{ord}_\infty(f) \geq -n \} \cup \{0\} \\ &= \left\{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \begin{array}{c} f \text{ may have a pole of order at most } n \text{ at } \infty \\ f \text{ has no pole at } p \in \mathbb{C} \end{array} \right\} \cup \{0\} \end{aligned}$$

Let

$$f(z) = P(z) \quad \text{with} \quad P(z) \in \mathbb{C}[z].$$

Since f may have a pole of order at most n at ∞, we require

$$\mathrm{ord}_\infty(f) = \mathrm{ord}_{w=0}\left(P(1/w)\right) = -\deg P \geq -n,$$

so $\deg P \leq n$. Thus $P(z) = a$ $(a \in \mathbb{C})$ and therefore $f(z) = a_0 + a_1 z + \cdots + a_n z^n$ $(a_i \in \mathbb{C})$. Hence

$$L(n[\infty]) = \mathrm{span}_{\mathbb{C}} \left\{1, z, z^2, \ldots, z^n\right\} \quad \text{and} \quad \ell(n[\infty]) = n+1.$$

(ii) Assume that $n < 0$. Then

$$\begin{aligned} &\{\, f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \mathrm{ord}_{p\in\mathbb{C}}(f) \geq 0;\ \mathrm{ord}_\infty(f) \geq -n \} \cup \{0\} \\ &= \left\{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \begin{array}{c} f \text{ has a zero of order at least } -n \text{ at } \infty \\ f \text{ has no pole at } p \in \mathbb{C} \end{array} \right\} \cup \{0\} \end{aligned}$$

Let

$$f(z) = P(z) \quad \text{with} \quad P(z) \in \mathbb{C}[z].$$

Since f has a zero of order at least n at ∞, we require

$$\operatorname{ord}_\infty(f) = \operatorname{ord}_{w=0}\left(P(1/w)\right) = -\deg P \geq -n,$$

so $\deg P \leq n$. Since $P(z) \in \mathbb{C}[z]$, $P(z) \equiv 0$ therefore $f(z) = 0$. Hence

$$L(n[\infty]) = \{0\} \quad \text{and} \quad \ell(n[\infty]) = 0.$$

□

Corollary 5. *For any divisor D on $\mathbb{CP}^1$,*

$$\ell(D) = \begin{cases} \deg D + 1, & \deg D \geq 0, \\ 0, & \deg D < 0. \end{cases}$$

Proof. Note that $D \sim (\deg D)[\infty]$ on $\mathbb{CP}^1$. □